

Question:

A silicious rock contains the mineral ZnS. To analyze for Zn, a sample of the rock is pulverized and treated with HCl to dissolve the ZnS (silicious matter is insoluable). Zinc is precipitated from solution by the addition of potassium ferrocyanide $K_4Fe(CN)_6$. After filtering, the precipitate is dried and weighed. The reactions which occur are $ZnS + 2HCl \rightarrow ZnCl_2 + H_2S$ $2ZnCl_2 + K_4Fe(CN)_6 \rightarrow Zn_2Fe(CN)_6 + 4KCl$ If a 2 gram sample of rock yields 0.969 gram of $Zn_2Fe(CN)_6$, what is the percentage of Zn in the sample? Atomic weight Zn = 65.4, molecular weight $Zn_2Fe(CN)_6 = 342.6$.

Answer Choices:

- A. 10 % Zn
- B. 22 % Zn
- C. 20 % Zn
- D. 19 % Zn
- E. 15 % Zn
- F. 18 % Zn
- G. 16 % Zn
- H. 14 % Zn
- I. 12 % Zn
- J. 25 % Zn

Reasoning + Answer:

<reasoning process here>

We are given :

- A 2 gram sample of rock yields 0.969 grams of $Zn_2Fe(CN)_6$.
- We are to find the percentage of Zn in the original rock sample .
- The reactions :
 1. $ZnS + 2HCl \rightarrow ZnCl_2 + H_2S$
 2. $2ZnCl_2 + K_4Fe(CN)_6 \rightarrow Zn_2Fe(CN)_6 + 4KCl$

We are told that the Zn in the rock is converted to $Zn_2Fe(CN)_6$ via the above reactions .

We are given :

- Molecular weight of $Zn_2Fe(CN)_6 = 342.6$ g/mol
- Atomic weight of Zn = 65.4 g/mol
- The precipitate is $Zn_2Fe(CN)_6$, which contains 2 Zn atoms .

So , in 1 mole of $Zn_2Fe(CN)_6$, there are 2 moles of Zn .

Step 1 : Find the mass of Zn in 0.969 g of $Zn_2Fe(CN)_6$.

Mass of Zn in the precipitate = (2 $\times$ atomic mass of Zn / molecular mass of $Zn_2Fe(CN)_6$) $\times$ mass of precipitate

= (2 $\times$ 65.4 / 342.6) $\times$ 0.969

Calculate :

2 $\times$ 65.4 = 130.8

130.8 / 342.6 $\approx$ 0.3818

0.3818 $\times$ 0.969 $\approx$?

0.3818 $\times$ 0.969 =

= 0.3818 $\times$ (1 - 0.031) $\approx$ 0.3818 - (0.3818 $\times$ 0.031)

= 0.3818 - 0.011836 $\approx$ 0.370

Better to compute directly :

0.3818 $\times$ 0.969

= (0.38 $\times$ 0.969) = 0.36822

(0.0018 $\times$ 0.969) $\approx$ 0.00174

Total $\approx$ 0.36996 $\approx$ 0.370 g

So , mass of Zn in the precipitate $\approx$ 0.370 g

This Zn came from the original 2 g sample .

Therefore , % Zn in the rock = (mass of Zn / mass of sample) $\times$ 100

= (0.370 / 2) $\times$ 100 = 18.5 %

Closest option to 18.5 % is :

F. 18% Zn

But let's compute more precisely .

Precise calculation :

(2 $\times$ 65.4) = 130.8

130.8 / 342.6 = ?

130.8 $\div$ 342.6 = ?